Communication Under Extreme Delay & Disruption

Delay-Tolerant Networking (DTN)

If you’re cut off, hold the message — and
forward it the moment you can.

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What we’ll cover

From an elevator glitch to an interplanetary network — in six steps.

When your text won’t send in an elevator

The message wasn’t lost. Your phone held it — and sent it when signal returned.

1

Signal drops

Send fails — no path right now.

2

The phone holds it

It queues the message and waits.

3

Signal returns → it sends

It forwards. Nothing was lost.

• Store · Carry · Forward

That instinct has a name: DTN.

Now scale that glitch to Mars

The same problem — blown up to the size of the solar system.

seconds

ELEVATOR — SIGNAL GONE BRIEFLY

3–22 min

MARS — ONE-WAY DELAY · LINK DOWN FOR HOURS

WHY PICK THE HARDEST CASE

Crack the Mars gap, and the deep sea, disaster zones, and remote areas come almost for free.

NASA

Standardized & flying

TCP assumes an unbroken path — right now. On Mars, that breaks.

One bundle. Earth → Mars. Five stages

Each stage = a question + the tech that answers it.

① **Why does TCP break?**

4 broken assumptions

② **Can’t we just hold it?**

Store-carry-forward

③ **Who’s responsible now?**

Custody transfer

④ **Reliable over minutes?**

LTP

⑤ **Which route, and when?**

Routing

It’s not slow internet — it’s broken physics

Four physical realities, none of which TCP was built for.

DELAY	DISRUPTION	CAPACITY	LOSS
Long & variable	Links keep dropping	Asymmetric	Physical, not congestion

Each one breaks an assumption TCP was built on.

Every TCP assumption breaks out here

Each reality collides with something TCP takes for granted.

TCP ASSUMES…	…AND IN SPACE IT COLLAPSES
A continuous path	Handshake never completes.
Short, stable RTT	ACKs too late — window can't grow.
Timeout = loss	Re-sends good data — wastes bandwidth.
Loss = congestion	Backs off when a bit just flipped by radiation.

Not a faster TCP — a different premise

Cut off? Hold it. Then forward it

Start from the obvious instinct — then close the gap it leaves.

Hold it, forward when the link returns

Just like your phone in the elevator. But if the node dies — who's responsible?

A node signs for delivery — then it's theirs

Like a courier depot relay. That sign-off is what separates DTN from plain caching.

DTN adds one layer: the bundle layer

It doesn’t replace the network — it overlays one layer and swaps transport per segment.

BUNDLE LAYER — UNIT OF DATA IS A BUNDLE, NOT A PACKET

SENDER
Earth

TCP

RELAY
Store & forward

LTP

RECEIVER
Mars

TCP can’t ride the space link. LTP can

LTP = Licklider Transmission Protocol, named for an internet pioneer.

IDEA 1 — RED & GREEN DATA

Split what must arrive from what can drop

RED — MUST ARRIVE	GREEN — BEST EFFORT

IDEA 2 — DEFERRED ACKS

Don’t wait on ACKs — match them to the RTT

Reliable transport, even at minute-scale delay.

RFC 5326 · Built for the space segment

It arrives — even with no end-to-end path

Filling the gaps, one at a time.

Store-and-carry

We don’t have to wait.

Custody

Responsibility carries forward.

LTP

Reliable on the space link.

One question left — which route, and when?

Know the future, or bet on who you meet

Two branches — by how predictable the network is.

PREDICTABLE — E.G. SPACE

CGR — Contact Graph Routing

Orbits are predictable, so it **precomputes when each link opens** from a contact plan.

Contact plan, computed ahead

UNPREDICTABLE — E.G. DISASTER NETS

Opportunistic routing

Spray copies onto whoever you run into.

Epidemic — spray to all: high delivery, huge overhead.

Spray-and-Wait — capped copies, balanced.

This isn’t theory. It’s flying

Already running in space — and reaching back down to Earth.

2008	Today	Ahead
DINET — first proven in space	ION (NASA JPL) runs on the ISS	LunaNet — DTN on the Moon

We didn’t just read about it. We ran it

We verified all five stages in running code — on a regular laptop.

1

Install ION-DTN

NASA JPL, open source.

github.com/nasa-jpl/ion-dtn

2

Fake the Mars delay

Built-in owltsim.

owltsim

3

Send into a dead link

Watch it arrive.

“Hello Mars”

Demo-proof Even if the live demo fails — a real execution-log capture backs it up.

Two nodes. One fake Mars delay

The smallest test that still shows a real end-to-end hop.

SENDER
"Earth"

600 S ONE-WAY →

RECEIVER
"Mars"

OWLTSIM INJECTS THE DELAY — MARS HEARS NOTHING FOR 600 S

• Store → hold → forward

At t = 0, "Hello Mars" is handed to Earth while the link is down — ION stores, holds, and forwards.

owltsim: honest about what it fakes

Does it mimic Mars, or just call sleep? Here's the line.

MODELS — THE TIMING

Holds every signal exactly 600 s. Models the **delay length** and the **link drop & return**.

DOES NOT MODEL — THE PHYSICS

No RF channel, antenna, or radiation bit errors. **Not a physics simulator.**

Why it's credible Store-carry-forward runs in ION's real code — not simulated. owltsim only feeds the delay.

Sent into a dead link — it arrived 600 s later

Receiver first, then send. Stored while disconnected → delivered intact.

```
t = 0      [earth]  "Hello Mars" stored — link DOWN
t = 0      [tcp]    connect() ... timeout — TCP would quit here
t = 600 s  [earth]  light-time elapsed → forwarding
t = 600 s  [mars]   bundle delivered: "Hello Mars"
t = 600 s  [mars]   custody OK — receiver accepted
```

TCP: FAILS

DTN: DELIVERS

TCP vs DTN — by the numbers

12 min is an assumed middle value (range 3–22).

UNDER A 12-MIN ONE-WAY DELAY	TCP	DTN
3-way handshake	~36 min to connect	Not needed
On a broken link	Dies, restarts, times out	Stored, delivered later
Delivery under disruption	**FAILS**	**100% (after delay)**

Not "faster" — a different premise

BDP: too much data “in flight”

Why TCP can’t just keep the pipe full across a 24-min round trip.

BDP = BANDWIDTH × RTT

≈ 180 MB

IN FLIGHT AT ONCE — 1 MBPS × 24-MIN ROUND TRIP

TCP

Can’t hold 180 MB un-ACKed → flow control paralyzed.

DTN

Stores per hop → never “all in flight.”

When NOT to use DTN

Knowing where it loses is as much a part of understanding it as where it wins.

THE COSTS

- **Heavy storage burden** — nodes hold data long.
- Congestion control still **immature.**

USE DTN

No continuous path **+** large delay/disruption.

USE TCP/IP

Stable + low latency — DTN there is a net loss.

What we proved — and where it stops

Honest about both the wins and the caveats.

WHAT WE ACCOMPLISHED

- ✓ Walked one bundle Earth→Mars through all five stages.
- ✓ Ran NASA's ION-DTN ourselves — delivered intact across a dead link.
- ✓ Quantified it — TCP fails under disruption, DTN delivers 100% after the delay.

LIMITATIONS

- A small, representative demo — two nodes, one injected delay, not a real RF link.
- owltsim models the timing, not the radio physics.
- No large-scale, security, or routing-at-scale testing.

Internet assumes a path open now. DTN assumes a path that’ll open someday.

Next time your text lands a second late, that’s the same instinct behind an interplanetary network.

Sources

Standards, implementations, and missions referenced throughout.

01 Bundle Protocol Version 7. IETF RFC 9171, 2022.

02 Licklider Transmission Protocol (LTP). IETF RFC 5326.

03 ION — Interplanetary Overlay Network. NASA JPL · github.com/nasa-jpl/ION-DTN.

04 DINET — first in-space DTN. NASA, 2008.

05 DSOC — Deep Space Optical Comms, Psyche. NASA, 2023.

06 CCSDS Bundle Protocol recommendations. CCSDS.